Dineth Wijesinghe

Site Reliability Engineering (SRE) Intern

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SUMMARY

Information Technology undergraduate at the University of Moratuwa seeking a Site Reliability Engineering (SRE) Internship. Practical experience applying Google SRE principles, architecting distributed observability pipelines with OpenTelemetry and Honeycomb, orchestrating containerized workloads on Kubernetes (K3s), and automating operational toil with Go, Python, and Bash. Proven ability in proactive incident triage, Root Cause Analysis (RCA), and engineering resilient CI/CD delivery pipelines for high-availability cloud infrastructure.

EDUCATION

BSc (Hons) in Information Technology **2024 – Present**

University of Moratuwa, Moratuwa, Sri Lanka

- **GPA :- 3.6 / 4.0**

G.C.E. Advanced Level **2020 – 2022**

H/North Kanumuldeniya National School, Hambantota, Sri Lanka

- Combined Mathematics | Physics | ICT · **Z-Score :- 1.7991**

KEY PROJECTS

CloudSight – Cloud Observability Platform (Group Project) Github Repo | Live Demo

▶ **Technologies :-** Node.js, Express.js, React, Python, Socket.IO, scikit-learn, Docker, AWS (EC2, RDS, ECR, Amplify), PostgreSQL, Redis, Nginx, CI/CD

CloudSight is an enterprise-grade, real-time infrastructure observability platform featuring a 6-model ML anomaly detection ensemble, automated incident lifecycle management, real-time WebSocket streaming, dynamic alerting, and PDF report generation.

Key features :-

- Architected a 6-model ML anomaly detection ensemble (IsolationForest, IQR rolling baselines, linear regression slope analysis, hard threshold rules, log-pattern spike detection, and stale-agent heartbeat monitoring) to surface real-time infrastructure degradations asynchronously.
- Engineered an automated incident lifecycle system with 3-tier severity triage (Warning, Critical, Emergency) and real-time WebSocket event streaming via Socket.IO, standardizing runbook escalation to reduce Mean Time to Triage (MTTT) by 40%.
- Architected push-to-deploy CI/CD pipeline (GitHub Actions + Amazon ECR) for zero-downtime rolling updates; configured private-VPC RDS with automated snapshots and Redis caching, offloading 40% of repetitive database query load.

AWS K3s OpenTelemetry Observability Lab Github Repo

▶ **Technologies :-** AWS EC2, K3s, Kubernetes, OpenTelemetry Collector, Honeycomb, OTLP, Docker, Linux, Bash

A production-grade SRE observability platform deployed on Kubernetes using OpenTelemetry and Honeycomb, simulating a real-world microservices environment with a focus on security, correctness, and operational reliability.

Key features :-

- Provisioned a production-grade K3s Kubernetes cluster on AWS EC2, deploying an OpenTelemetry Demo microservices mesh with AWS Security Group network isolation, namespace CPU/memory resource quotas, and health probes for self-healing.
- Engineered a unified OpenTelemetry Collector pipeline exporting telemetry over secure OTLP/HTTPS to Honeycomb, eliminating redundant demo-only exporters to reduce memory footprint by 35% on resource-constrained nodes.
- Configured Honeycomb datasets with p95/p99 latency waterfall views to monitor SLIs, and integrated Honeycomb MCP (Model Context Protocol) to query distributed traces and diagnose service bottlenecks via AI-assisted natural language.

Tunnix – Self-Hosted Ingress & Developer Tunneling Platform Github Repo | Live Demo

▶ **Technologies :-** Go, gorilla/websocket, Bun, Docker, AWS (EC2, CloudFront, S3, Amplify), Nginx, SQLite (WAL), TypeScript, React, Bash

Tunnix is a high-performance, self-hosted developer platform that securely exposes local HTTP servers to the internet using WebSocket frame multiplexing, featuring automated email OTP authentication, a React SPA dashboard, and an administrative governance suite.

Key features :-

- Engineered a production Go ingress gateway (`gorilla/websocket`) multiplexing thousands of concurrent HTTP requests over bi-directional WebSocket frames, sustaining sub-50ms round-trip latency with cross-platform CLI agent support (Linux, macOS, Windows).
- Automated zero-touch SSL lifecycle management (Let's Encrypt), AWS CloudFront edge routing, and dynamic Nginx configuration via idempotent systemd deployment scripts – eliminating recurring certificate-renewal operational toil.
- Hardened the control plane with a 4-tier auth hierarchy (Email OTP → Agent Tokens → Access/Refresh JWTs → Tunnel Grant JWTs), sliding-window rate limiting, and an admin governance suite with emergency disconnect and SQLite WAL-mode audit logging.

TECHNICAL SKILLS

- **Languages :-** Go (Golang), Python, Bash/Shell, TypeScript, SQL
- **Observability & Telemetry :-** OpenTelemetry (OTel), Honeycomb, Prometheus, Grafana, Distributed Tracing, Alerting
- **Cloud, IaC & Orchestration :-** AWS (EC2, ECR, RDS, S3, CloudFront), Kubernetes (K3s), Terraform, Docker, Nginx
- **Systems & CI/CD :-** Linux (Systemd, Networking, cgroups), GitHub Actions, Zero-Downtime Rollouts, TCP/IP

CERTIFICATIONS & COURSES

- LFS158: Introduction to Kubernetes – The Linux Foundation · Architecture, Deployments & Pod Networking *(Ongoing)*
- LFS101: Introduction to Linux – The Linux Foundation · Linux Administration, Shell Scripting & System Processes
- CCNA (Cisco Certified Network Associate) – Enterprise Networking, TCP/IP, Routing & Switching *(Ongoing)*
- AWS Cloud Essentials – Core AWS Infrastructure · VPC, IAM, EC2, RDS, and Cloud Architecture *(Ongoing)*

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

- **Batch Representative** | **Faculty of Information Technology, University of Moratuwa** 2026 – Present
- **Programme Committee Member** | **Riseup Mora 2026, University of Moratuwa** 2026
- **Senior Prefect** | **H/North Kanumuldeniya National School** 2019 – 2020

REFERENCES

Mr. Indika Wimalasuriya
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